William Hogg

Project Portfolio:
WilliamWHogg.com

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Education

M.S. in Mechanical Engineering - GPA: 4.0 ***May 2026***

Washington State University, Pullman, WA

Relevant Coursework: Design for Additive Manufacturing, Topics in Additive Manufacturing, Topology Optimization, Cryogenics

B.S. in Mechanical Engineering - GPA: 4.0 ***May 2025***

Washington State University, Pullman, WA

Minors: Material Science, Mathematics

Relevant Coursework: Mechanics of Solids, Metallic Materials, Material Characterization, Mathematical Optimization

Experience

WSU - Thermal and Fluids Laboratory - Teaching Assistant ***Jan 2026 – May 2026***

- Guided 16 students on setup, uncertainty analysis, interpretation and reporting of thermal-fluids experiments

W.M. Keck Biomedical Materials Lab - Research Assistant ***Aug 2025 – Dec 2025***

- Developed, planned, and executed laser powder bed fusion (LPBF) manufacturing processes for brittle titanium-ceramic composites using batch Bayesian optimization and structured experimentation, reducing cracking and enabling repeatable builds on a 3DS DMP 200 system
- Developed MATLAB tools to automate powder size characterization and mechanical test data processing, improving consistency and reducing analysis time by ~5×
- Maintained and repaired LPBF system and trained researchers on machine operation and laboratory safety procedures

W.M. Keck Biomedical Materials Lab - Undergraduate Researcher ***Apr 2023 – Aug 2025***

- Designed and implemented corrosion-fatigue test protocols for additively manufactured titanium alloys and conducted failure-mode characterization resulting in a first-author publication
- Programmed CAM (Fusion 360) for CNC milling and turning of miniaturized hip joint implant components constructed from additively manufactured alloys and medical grade polymer
- Performed mechanical and microstructural characterization of additively manufactured alloys including tensile, fatigue, hardness, optical microscopy, SEM and fractography

Exotic Metals Forming - Project Engineer Intern ***Summer 2022***

- Conducted root-cause failure investigations on titanium superplastic forming; identified process and tooling contributors and updated work instructions and drawings in collaboration with manufacturing, quality, and tooling engineers
- Reviewed and revised production drawings and GD&T for formed, riveted, laser cut, and welded aerospace assemblies, improving pass rate and manufacturability
- Designed and implemented work holding and transfer fixture to support roller seam welding operations, improving ergonomics, safety and repeatability on a 90 lb component

Frank Innovation Zone (FIZ) - Instructor/Technician ***Jan 2022 – Apr 2023***

- Trained 100+ students and maintained a 3,000+ sq ft makerspace, supporting safe operation of shop equipment

Eagle Scout - Troop 624 ***2014 – 2021***

Featured Projects - Details on WilliamWHogg.com

Solo-String Art Printer - Designed custom 3D printed polymer cycloidal drive actuator for precise rotation ***Ongoing***

Team-Classroom Injection Molding Machine - Defined design requirements, selected polymer and tooling materials ***Apr 2025***

Team-Deployable Multi-Actuator Wheelchair Lift - Defined design requirements, led electronics and controls team ***Dec 2024***

Skills & Abilities

Design & Analysis: SOLIDWORKS, 3DXpert, GD&T, COMSOL, MATLAB(data processing and visualization), Python

Manufacturing: Additive Manufacturing (LPBF, FDM, SLA, L-DED, WA-DED), CNC Machining, Sheet Metal Forming, Welding

Characterization: Mechanical Testing (Fatigue, Fracture, Tension/Compression, Hardness), Optical Microscopy, SEM/EDS

Publications and Honors

"Contamination of 316L Powder with Ti6Al4V During Laser Powder Bed Fusion" *Virtual and Physical Prototyping* **Feb 2026**

"Corrosion-Fatigue of Additively Manufactured Ti6Al4V" *Journal of the Mechanical Behavior of Biomedical Materials* ***Dec 2025***

VCEA-Outstanding Junior in Mechanical Engineering ***April 2024***

"Fatigue Behavior of Additively Manufactured Ti3Al2V Alloy," *Materials Science in Additive Manufacturing* ***Sep 2023***